INFYSLATE · SUPPLY CHAIN AI

The Infyslate Control Tower

> One AI-native command center for stock, logistics, documents & money — Business Plan & Pitch Demo Narrative

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Document: 1 of 3 — Business Plan
Companion documents: Technical Architecture & Build Plan · UI/UX & Demo Design
Prepared for: Infyslate founder & pitch audiences (investors, design partners)
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Status: Pre-build planning document — demo is built on sample data, not live customer data

1. Executive Summary

> Infyslate is building one AI control tower that watches a company's stock, shipments, paperwork and money together, instead of asking operators to stitch together five disconnected tools and a dozen spreadsheets. The pitch demo shows seven connected modules running on realistic sample data, with a single AI alerts feed that proves the core thesis: most supply chain losses are not a forecasting problem or a logistics problem in isolation — they are a *visibility* problem that only shows up when you look at inventory, transfers, documents and reconciliation side by side.

Today the founder has already scoped three of the seven modules (3PL management, Replenishment & Planning, and Document AI). This plan adds four more — Stock Transfer Optimization, Salvage/Liquidation Optimization, Out-of-Stock Prediction, and Reconciliation with AI Flagging — and ties all seven into one demonstrable product: the **Control Tower**. The purpose of this document is to give each module a clear business case, target user, demo narrative and set of constraints, so that the technical and UI plans (documents 2 and 3) can be built against a single, consistent story.

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AI MODULES, ONE DATA SPINE	UNIFIED ALERTS FEED	END-TO-END DEMO	LIVE CUSTOMER DATA NEEDED

2. The Problem

Mid-market brands, distributors and the 3PLs that serve them run on a patchwork of WMS, ERP add-ons, spreadsheets and email. Four failure patterns repeat across almost every operator conversation in this space:

Pattern	What actually happens
Stockouts surprise everyone	Demand shifts faster than weekly reorder cycles; by the time a human notices a SKU is low, it is already out, and the lost sale is invisible in most reporting.
Cash sits in dead stock	Slow-moving and obsolete inventory quietly accumulates because no one is scored or incentivized to clear it until a year-end write-off forces the conversation.
Paperwork eats operator time	POs, invoices, bills of lading and goods-receipt notes are matched by hand; a transposed quantity or price is caught weeks later, if at all.
Warehouses hoard or starve	One location sits on six months of a SKU while another stocks out of the same SKU, because transfer decisions are reactive, manual and rarely made on real cost data.

Point solutions exist for each pattern individually (demand planning tools, liquidation marketplaces, invoice-matching software, warehouse management add-ons). What does not exist at the price point and simplicity this segment needs is one view that correlates all four — so that a reconciliation mismatch, a developing stockout and an excess-inventory position on the same SKU are seen as one story, not three unrelated alerts in three different logins.

3. The Solution: One Control Tower, Seven Modules

The Infyslate Control Tower is a single home screen backed by seven purpose-built modules that all read and write to the same underlying data spine (SKU, Location, Vendor, Carrier, Order, Shipment, Invoice). Every module produces structured **AI flags** — a consistent "what, why, how much, what to do" object — that surface both inside the module and on the Control Tower home feed. This is the single most important design decision in the whole product: the demo is convincing not because any one module is novel, but because the *same screen* shows a stockout risk, the excess inventory at a sister warehouse that could cover it, and the invoice mismatch that explains why nobody reordered on time.

Module	Status	One-line promise	Primary KPI shown in demo
3PL Control	Scoped	See every carrier's cost and SLA performance in one scorecard	On-time % & cost/shipment by carrier
Replenishment & Planning (R&P)	Scoped	Know what to reorder, how much, and when — automatically	Service level at target safety stock
Document AI	Scoped	Turn POs, invoices and BOLs into clean structured data in seconds	Straight-through extraction rate
Stock Transfer Optimization	Added	Move inventory between locations before you buy more of it	$ saved vs. new purchase
Salvage / Liquidation Optimization	Added	Turn aging, excess stock into recovered cash before it's written off	$ recovered & days-of-inventory reduced
Out-of-Stock Prediction	Added	See a stockout coming 2–4 weeks out, not the day it happens	Stockouts avoided / lead-time-to-flag
Reconciliation + AI Flagging	Added	Auto-match PO → receipt → invoice and flag every exception	% auto-matched, exceptions caught

"Scoped" modules are the three the founder had already defined before this plan (3PL, R&P, Document AI); "Added" modules are the four introduced in this plan to complete a full control tower story. All seven carry equal weight in the demo — the labels above are for traceability against the original idea only, and should not appear in the product itself.

4. Module-by-Module Business Case

4.1 3PL Control

Problem: Shippers using multiple 3PLs/carriers have no single scorecard for cost, speed and reliability, so bad carriers keep getting volume out of inertia.

Who it's for: Ops/logistics managers at brands and distributors running 2–6 carrier or 3PL relationships.

What the demo shows: A carrier scorecard (cost per shipment, on-time %, damage/claims rate) across 4–5 sample carriers, a rate-shopping recommendation for an upcoming shipment, and an SLA-breach alert feeding the Control Tower.

Business value / KPIs: Freight cost reduction (typically framed as a %), on-time delivery improvement, time saved on manual carrier comparison.

CONSTRAINTS FOR THE DEMO BUILD

Only 4–5 synthetic carriers with pre-set rate cards and performance histories; no live carrier API or real rate shopping; "recommendation" is a deterministic scoring formula, not a live auction; demo must not imply integration with any named real carrier.

4.2 Replenishment & Planning (R&P)

Problem: Reorder points are set once and forgotten; planners either over-buy (tying up cash) or under-buy (causing the stockouts Section 4.6 is built to catch).

Who it's for: Inventory/demand planners at brands with 50–5,000 active SKUs.

What the demo shows: A SKU list with live reorder recommendations, a demand forecast chart per SKU, an adjustable service-level/safety-stock slider, and a one-click suggested purchase order.

Business value / KPIs: Target service level achieved with lower average inventory; fewer emergency POs; planner hours saved per week.

CONSTRAINTS FOR THE DEMO BUILD

Forecasting method must be simple and explainable (moving average / exponential smoothing), not a black box, since investors and design partners will ask "how does it know"; demo dataset limited to ~60–100 SKUs across a handful of categories so the forecast charts stay readable; no true multi-echelon optimization in v1 — single-location reorder logic per SKU, extended later.

4.3 Document AI

Problem: POs, invoices, bills of lading and goods-receipt notes arrive as PDFs/scans/emails and are keyed in by hand, which is slow and error-prone, and is also the raw material Reconciliation (4.7) needs.

Who it's for: AP/ops clerks and controllers who currently re-type supplier documents.

What the demo shows: A document inbox of sample invoices/BOLs/POs, a side-by-side extracted-fields viewer with confidence highlighting, and automatic hand-off of extracted data into Reconciliation.

Business value / KPIs: Straight-through processing rate, minutes saved per document, error reduction vs. manual entry.

CONSTRAINTS FOR THE DEMO BUILD

Demo uses a fixed library of 10–15 synthetic documents (generated as PDFs, not scans of real supplier paperwork) so extraction is reliable on stage; genuinely run OCR/LLM extraction live if feasible within the build budget, but ship a cached/offline fallback result for each sample document so the demo never depends on live model latency or an internet connection on pitch day.

4.4 Stock Transfer Optimization

Problem: One warehouse is sitting on excess of a SKU while another is about to stock out of the same SKU, and nobody connects the two because transfer decisions require manually comparing inventory positions across locations.

Who it's for: Multi-location retailers, distributors and 3PLs with 3+ warehouses or stores.

What the demo shows: A network view of surplus vs. deficit locations for the same SKUs, a ranked list of recommended transfers with cost/benefit vs. buying new stock, and a one-click "approve transfer" that clears the matching Out-of-Stock alert.

Business value / KPIs: $ of avoided purchases, inventory days-of-supply rebalanced, transfers executed per week.

CONSTRAINTS FOR THE DEMO BUILD

Transfer cost model uses a simplified distance/weight-based estimate, not real freight rates; location count capped at 5–8 for a legible network view; must visibly tie back to the Out-of-Stock Prediction and Salvage modules on the same SKU to land the "one data spine" story.

4.5 Salvage / Liquidation Optimization

Problem: Slow-moving and obsolete stock is usually only addressed at year-end write-off time, by which point most of its recoverable value is gone.

Who it's for: Inventory/finance managers tracking aging and obsolete stock, and brands with seasonal or perishable-adjacent goods.

What the demo shows: A ranked "at-risk inventory" list scored by age, days-of-supply, margin erosion and channel fit; a recommended disposition per SKU (discount channel, liquidation partner, bundle, write-off); and an estimated $ recovered if acted on this week vs. next quarter.

Business value / KPIs: $ recovered, reduction in average inventory age, reduction in write-off rate.

CONSTRAINTS FOR THE DEMO BUILD

Scoring formula and recovery-value estimates must be clearly labeled as illustrative/sample-data outputs, not audited financial figures; no live connection to real liquidation marketplaces; disposition "recommendation" is rules + scoring, not a live marketplace price feed.

4.6 Out-of-Stock (OOS) Prediction

Problem: Most teams learn about a stockout the day it happens (or from a customer complaint), when a 2–4 week warning would have left time to reorder or transfer stock.

Who it's for: Planners and category managers accountable for sell-through and fill rate.

What the demo shows: A risk-tiered SKU list (red/amber/green) with a days-to-stockout countdown, a demand-vs-supply chart per SKU, and an explainability panel showing exactly why a SKU was flagged (sales velocity up, incoming PO delayed, no transfer available).

Business value / KPIs: Stockouts avoided, average lead time between flag and stockout, revenue at risk surfaced.

CONSTRAINTS FOR THE DEMO BUILD

Sample data must be scripted so a known subset of SKUs is guaranteed to cross into amber/red risk during the demo window (see Technical Plan §5) — the demo cannot depend on a forecast "happening" to be interesting; explainability panel is mandatory, not optional, since "trust the AI" is the hardest sell in this category.

4.7 Reconciliation with AI Flagging

Problem: Three-way matching (purchase order vs. goods receipt vs. invoice) is done manually or with brittle exact-match rules, so mismatches in quantity, price or dates are missed until they show up as a cash problem.

Who it's for: AP/controllers and ops managers who sign off on supplier payments.

What the demo shows: A 3-way match table across sample POs/receipts/invoices with automatic fuzzy matching, severity-coded exception flags (price variance, quantity variance, duplicate invoice, missing receipt), and a drill-down showing exactly which fields disagree and by how much.

Business value / KPIs: % auto-matched without human review, $ of discrepancies caught before payment, hours saved per AP cycle.

CONSTRAINTS FOR THE DEMO BUILD

At least 3–4 of the sample document sets must be deliberately seeded with a mismatch (wrong quantity, wrong unit price, duplicate invoice number) so the "AI flagging everything" promise is visibly, reliably true every time the demo runs; matching logic should be transparent (rule thresholds + fuzzy string/number matching), not framed as unexplainable AI.

5. Demo Narrative (the pitch walkthrough)

A single scripted path through the Control Tower, designed to run in under ten minutes and to always land the same three beats: *one data spine, not five tools*; *the AI explains itself*; *a dollar number appears*. Each demo must open on the Control Tower home screen, not a module, so the first thing an investor sees is the unified alerts feed.

Beat	Screen	What the presenter says / shows
1	Control Tower Home	"This is one dashboard. It already found 6 issues across stock, shipping and paperwork — let's look at the biggest one."
2	Out-of-Stock Prediction	Click the top alert → shows a SKU about to stock out in 9 days, with the explainability panel open.
3	Stock Transfer Optimization	"Before we reorder — Warehouse C already has 400 units sitting idle." One click approves a transfer, alert clears.
4	Reconciliation	"Here's why this SKU was under-ordered in the first place" — shows the invoice/PO mismatch that delayed the last purchase order.
5	Salvage Optimization	Pivot to a second SKU: aging inventory with a recommended disposition and a recovered-value number.
6	R&P + 3PL + Document AI	Quick tour of the remaining three modules to show depth, closing on the carrier scorecard and document inbox.
7	Control Tower Home	Return to the home screen: "Six issues, one morning, one login."

6. Target Customers & Market Framing

Primary beachhead: mid-market D2C/CPG brands and regional distributors running 1–10 warehouses or 3PL relationships, with an inventory/ops lead but no dedicated supply-chain planning team or budget for enterprise suites (SAP IBP, Blue Yonder, o9). Secondary: small-to-mid 3PLs themselves, who need a control-tower view across the multiple brands they serve. The category — AI-driven supply chain control towers — is an active area of vendor and analyst attention through 2026, with incumbents moving from static dashboards toward more autonomous, AI-orchestrated versions; this plan's job is to make that same positioning credible at a price point and setup time this segment can actually afford. (Any market-size figures used on pitch slides should be sourced fresh from an analyst report at fundraising time rather than asserted here — this document intentionally avoids inventing a TAM number.)

7. Competitive Landscape (brief)

Category	Examples of the type	Where Infyslate differs
Enterprise control towers	SAP/Blue Yonder/o9-style suites, large systems integrator offerings	Months-long implementation, enterprise pricing; Infyslate targets same-week setup for mid-market.
Point solutions	Standalone demand-planning tools, excess-inventory/liquidation tools, invoice-matching tools	Each solves one slice; none correlates stock, transfers, documents and reconciliation in one view.
Spreadsheets & email	The real incumbent for most of this segment	Any structured, AI-assisted alternative is a significant upgrade; the bar to beat is low but trust must be earned with explainability.

8. Go-to-Market & Pricing Hypothesis

Land with one or two modules that map to the loudest pain (typically Out-of-Stock Prediction or Reconciliation), expand into the full Control Tower as trust is established. Hypothesis: per-location or per-SKU-band subscription pricing with the Control Tower view included as the retention hook once two or more modules are active — validate with design partners before quoting numbers externally.

9. Roadmap

Phase	Goal	Scope
Phase 0 — Demo	Pitch-ready	All 7 modules on synthetic sample data, no real integrations, built solo via Claude Code (this plan).
Phase 1 — Design-partner pilot	Validate with 1 real operator	1–2 highest-value modules connected to one real data source (CSV/API import), rest stay sample-data for context.
Phase 2 — Production	Multi-tenant product	Real integrations (ERP/WMS/carrier APIs), auth & data isolation, audited financial figures, SOC2-track security posture.

10. Risks & Constraints (company-wide)

Risk / constraint	Mitigation for the demo phase
Solo-founder build bandwidth	Scope every module to its demo-only constraint list (Section 4); technical plan sequences the build into small, Claude-Code-sized sprints.
"Is this real AI or smoke and mirrors?"	Every module's logic is explainable and stated honestly in diligence conversations as rules/statistics/light ML on sample data, with a clear roadmap to more sophisticated models in production.
Demo-day technical failure	Demo must run fully offline on a laptop with no dependency on live external APIs or Wi-Fi (see Technical Plan §9).
Overclaiming to investors	Any $ saved / $ recovered figures shown in the demo are explicitly labeled "illustrative, sample data" on-screen and in the pitch script.
Data privacy perception	No real customer, supplier or personal data anywhere in the demo build; this is stated directly in the pitch, not left ambiguous.

11. Glossary

Term	Meaning in this plan
Control Tower	The single home dashboard that aggregates AI flags from all seven modules.
R&P	Replenishment & Planning — reorder point / forecast / safety stock module.
3-way match	Reconciling a purchase order, its goods-receipt note, and the supplier invoice for the same transaction.
Days of supply / inventory age	How many days current stock would last at recent sell-through, and how long it has sat since receipt.
Safety stock	Buffer inventory held to protect a target service level against demand and lead-time variability.

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Sources consulted while preparing this plan (industry framing and terminology only — no figures from these sources are quoted as fact in this document): Tredence, "Supply Chain Control Towers in 2026" · Cognizant, "Harnessing Control Towers for Global Supply Chain Excellence" · Unstract, "A 2026 Guide to AI-Powered Logistics Document Processing" · SAP Insider, "Challenges of Excess Inventory Management" · SolveXia, "What is AI Reconciliation?" · Aston/Kingston University research, "A Machine Learning Approach to Inventory Stockout Prediction."